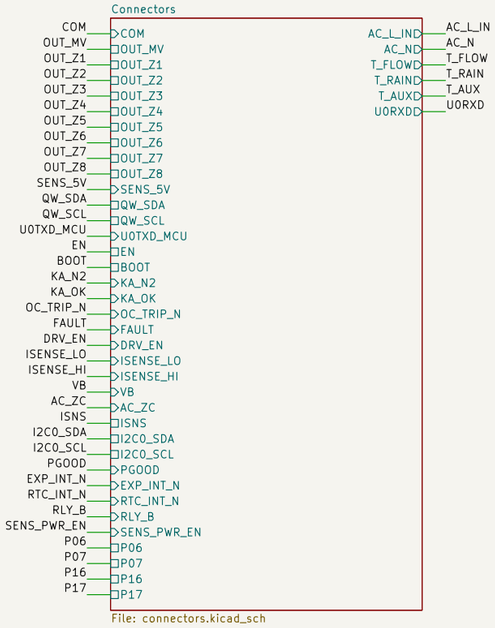
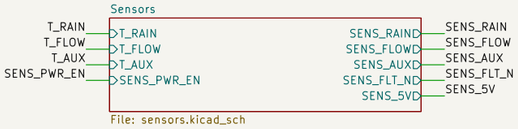
ZoneFlow rev A: block diagram



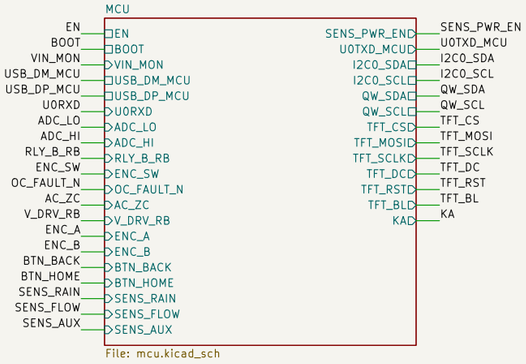




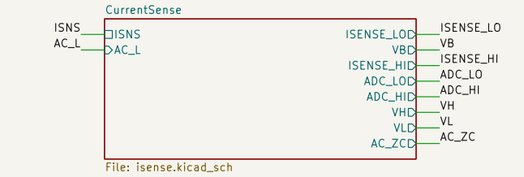


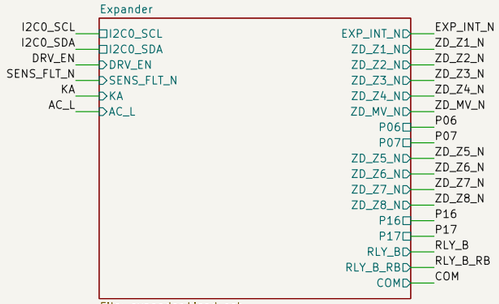


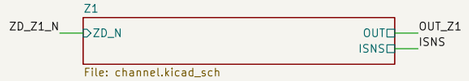


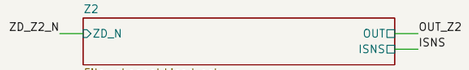


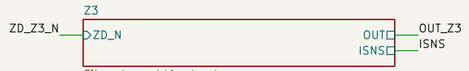


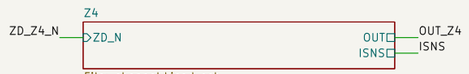






















Each box is one schematic sheet. A pin name is the net name: pins with the same label are joined.
Supply rails are not drawn here. They reach every sheet through power symbols:

GND	circuit ground. One transformer leg (AC_N behind F1), USB ground and sensor ground.
VIN_DC	rectified 24 VAC, 26 to 31 V, 44 V at 32 VAC with no load
V_BUCK	5.28 V from the LM5164, present only with 24 VAC. Also feeds the relay coil and the sensor supply.
V5	V_BUCK or VBUS through a Schottky diode each: about 4.95 V from 24 VAC, 4.1 to 4.9 V from USB
3V3	logic rail, 3.20 to 3.40 V
V_DRV	3V3 behind Q3: the opto LED supply, on only while the fail-safe circuit allows drive
VBUS	USB-C VBUS

The nine output channels Z1 to Z8 and MV are nine instances of one sheet, channel.kicad_sch.
References there are 101 to 111 style: the first digit is the channel (9 = master valve), so R301 is
the opto LED resistor of zone 3 and Q901 the master valve triac.

BUILD VARIANTS. The base design is the 8-zone build. KiCad variant "4-zone" does not fit
instances Z5 to Z8 (44 parts), J3 and R28. C52 and C53 are footprints only on both builds.

HOW A VALVE TURNS ON. Firmware toggles KA. Detector A (Failsafe sheet) raises DRV_EN, which releases
the expander from reset and switches V_DRV on. Detector B (Expander sheet) turns on Q2. Firmware then
sets RLY_ON (relay K1 feeds COM) and pulls a ZD_x_N pin low: the opto LED of that channel conducts and
its triac fires from its own T2. Valve current returns through ISNS and the shunt on the CurrentSense
sheet. When KA stops, both detectors let go within about a second, each on its own.

Connectivity is taken from hardware/netlist.yaml. The build fails unless the netlist KiCad extracts
from these sheets equals it pin for pin. This design has never been built.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /
File: zoneflow-a.kicad_sch

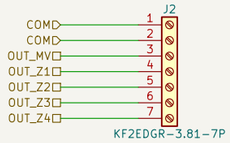
Title: ZoneFlow irrigation controller, revision A

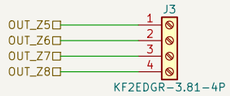
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/20

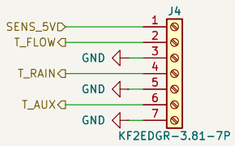
Field terminals: pluggable 3.81 mm headers



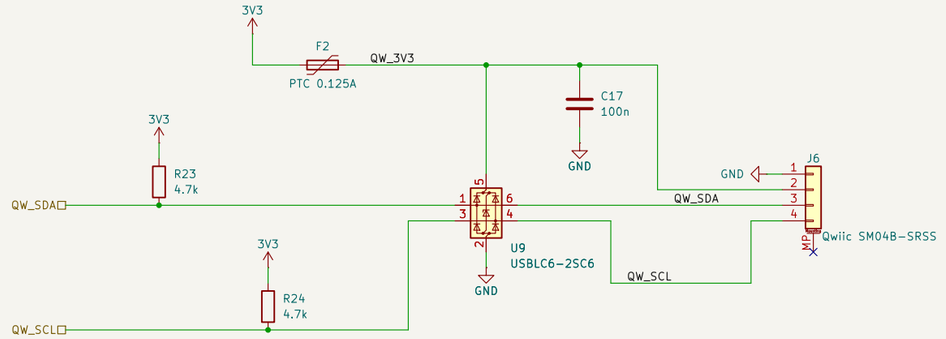
Terminal order along the bottom edge, left to right as the installer sees it:
J1 24V- 24V- transformer; pin 2 (AC_N) becomes circuit GND
J2 COM COM MV 1 2 3 4
J3 5 6 7 8 8-zone build only (KiCad variant "4-zone": not fitted)
J4 5V FLOW GND RAIN GND AUX GND
Headers KF2EDGR-3.81, plugs KF2EDGK-3.81 supplied loose. 300 V, 8 A.
Mark terminal 2 of J1 on the silkscreen.





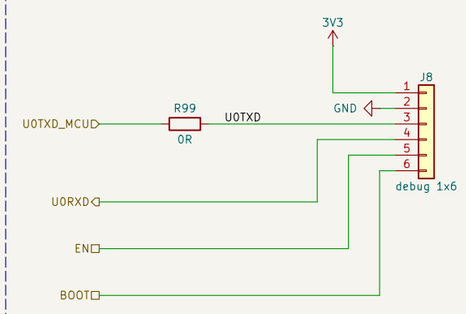


Qwiic expansion port on its own bus, I2C1



QWIIC (section 16.2). F2 holds 0.125 A at 20 C and 0.08 A at 60 C: the port supplies 80 mA.
R23, R24 4.7k. U9 clamps the bus lines; place it at J6. Pin order 1 GND, 2 3V3, 3 SDA, 4 SCL.

Factory and debug header (pads only)

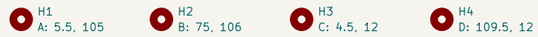


TEST PADS: 1.5 mm, top side. Ground around TP8.
No pad may sit in the fuse well.

Test pads

TP1	VIN_DC	TP11	FAULT	TP21	EXP_INT_N
TP2	V_BUCK	TP12	DRV_EN	TP22	COM
TP3	V5	TP13	ISENSE_LO	TP23	RTC_INT_N
TP4	3V3	TP14	ISENSE_HI	TP24	RLY_B
TP5	GND	TP15	VB	TP25	SENS_PWR_EN
TP6	GND	TP16	AC_ZC	TP26	P06
TP7	V_DRV	TP17	ISNS	TP27	P07
TP8	KA_N2	TP18	I2C0_SDA	TP28	P16
TP9	KA_OK	TP19	I2C0_SCL	TP29	P17
TP10	OC_TRIP_N	TP20	PGOOD	TP30	SENS_5V

Mounting holes (mechanical contract)



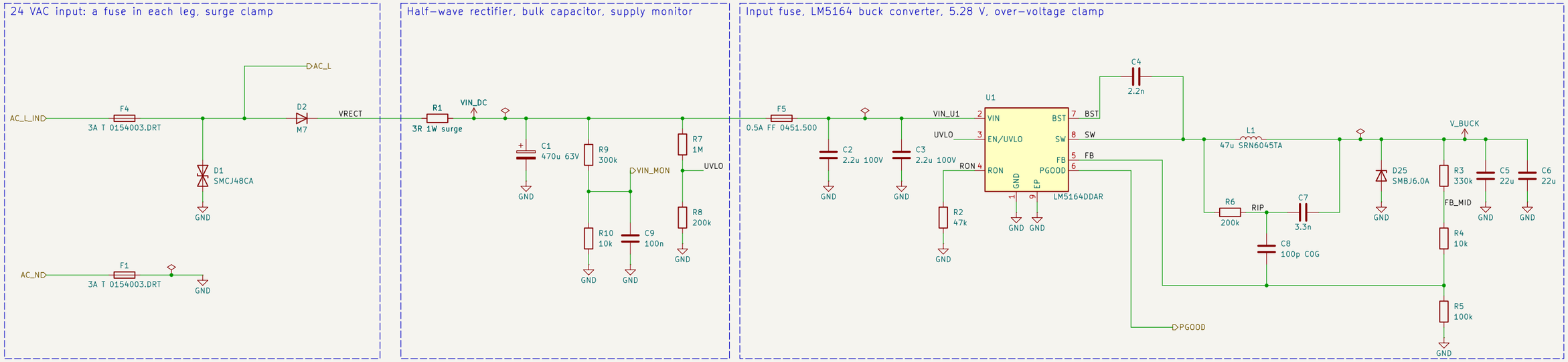
H1 to H4: 3.4 mm holes for M3,
at the positions of holes A to D.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Connectors/
File: connectors.kicad_sch

Title: Connectors, test pads and mounting holes

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/20



24 VAC INPUT AND LOGIC SUPPLY (hardware design section 4).

F4, F1 3 A time-lag fuses in holders, one in each transformer leg (section 5.3 says why two).
Terminal 2, AC_N, becomes circuit GND behind F1. Both holders sit in the fuse well behind J1.
D1 SMCJ48CA: stands off 48 V, above the 45.3 V peak of 32 VAC. Clamps at 77.4 V.
D2, R1 Half-wave rectifier. R1 is the surge-rated SR2512JK (5 %), not the FK (1 %) part:
it takes 237 to 385 W for about 1 ms when the board is plugged in at the mains peak.
C1 470 uF 63 V, through-hole, on the bottom side lying down. VIN_DC is 26 to 31 V at 24 VAC
and 44 V at 32 VAC with no load. Ripple 0.16 A typical, 0.48 A at the peak corner.
R9, R10 VIN_MON = VIN_DC / 31 to GPIO3: about 0.97 V at 24 VAC, 1.43 V at 32 VAC.
R7, R8 Under-voltage lockout of U1: on at 9.0 V, off at 8.4 V. On VIN_DC, before F5.

U1 LM5164, 100 V synchronous buck (TI SNVSAU4D). It survives the 77.4 V clamp level.
V_BUCK = 1.2 V x (1 + (R3 + R4) / R5) = 5.28 V, 5.16 to 5.40 V over tolerance.
R2 47k sets the on-time: 281 kHz.
F5, D25 For a switch of U1 that fails short (section 4.8). D25, SMBJ6.0A, holds V_BUCK under
10 V and F5, 0.5 A very fast, opens. F5 is soldered: it opens only when U1 has failed.
Do not fit a larger rating. Its 0.42 ohm also limits the fault current.
C4 2.2 nF bootstrap. The datasheet fixes this value; a larger one damages the part.
R6, C7, C8 Type 3 ripple injection: 24 mV at FB at 34 V in, 13 mV at 10 V.
L1 47 uH, Isat 2.0 A. Ripple 0.34 A, peak 0.99 A against a 1.25 A current limit.
Load on V_BUCK: 0.28 A typical, 0.78 A peak (section 4.2). It also feeds the relay coil and
the sensor supply directly, and V5 through D3 on the USB sheet.

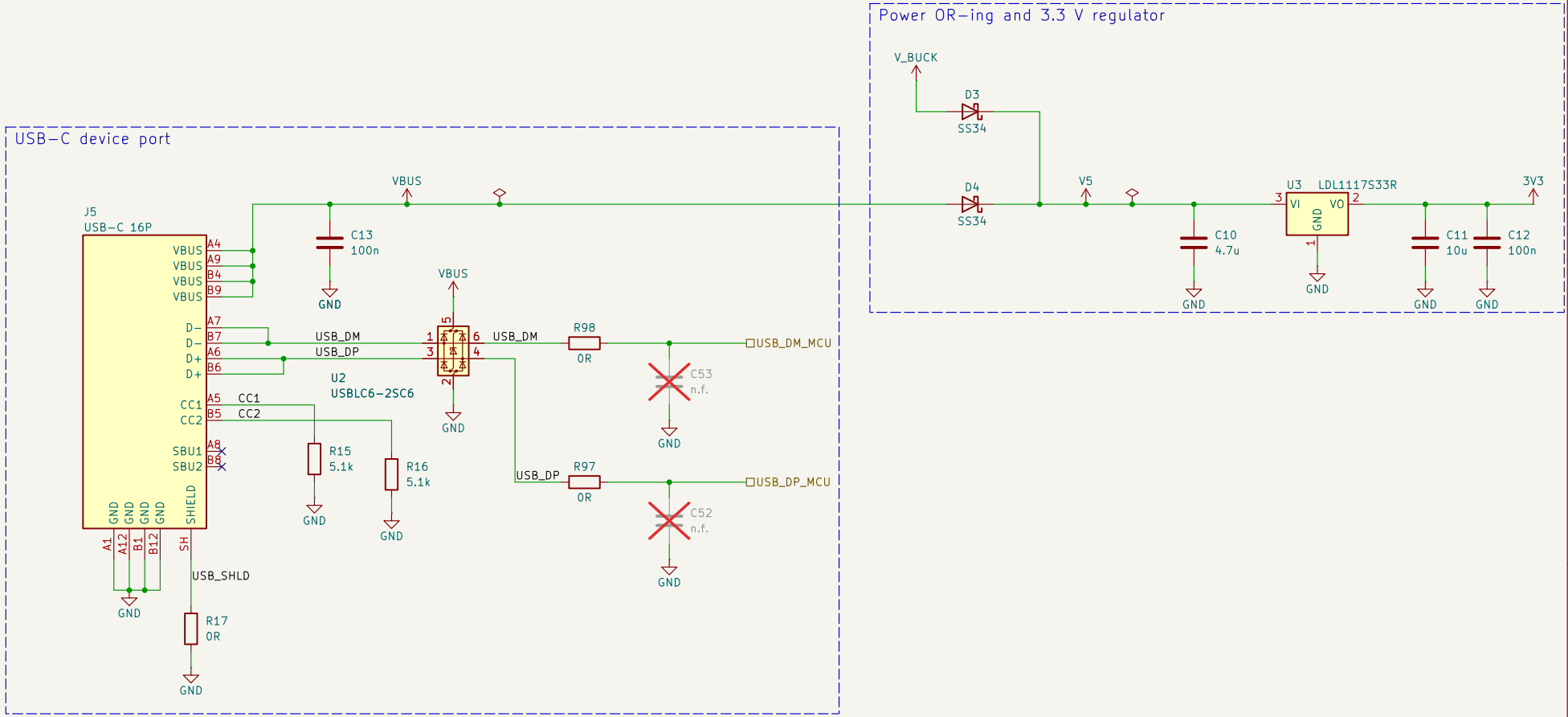
LAYOUT: C2 and C3 at U1 pins 2 and 1 with the shortest loop, behind F5. Keep the SW node small and away
from the current-sense parts. PWR_FLAG marks nets that are fed through a passive part.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Power/
File: power.kicad_sch

Title: 24 VAC input and logic supply

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/20



USB-C AND POWER OR-ING (hardware design section 5).
J5 is a USB 2.0 device port. R15 and R16, 5.1k on CC1 and CC2, mark the board as a device.
U2 clamps D+, D- and VBUS: place it at J5. R97 and R98 are 0R, C52 and C53 are not fitted:
series and shunt footprints for EMC work, placed within a few millimetres of the module pins.
D3 (from V_BUCK) and D4 (from VBUS) OR the two sources onto V5: about 4.95 V from 24 VAC,
4.1 to 4.9 V from USB. Neither source can feed the other. USB alone runs the logic, not the valves.
U3 LDL1117S33R: 3V3 is 3.20 to 3.40 V. Load 162 mA typical, 498 mA peak. Give the tab copper.
USB ground is circuit ground, which is one leg of the 24 VAC transformer: read section 5.3
and the safety page before connecting a mains-powered computer.

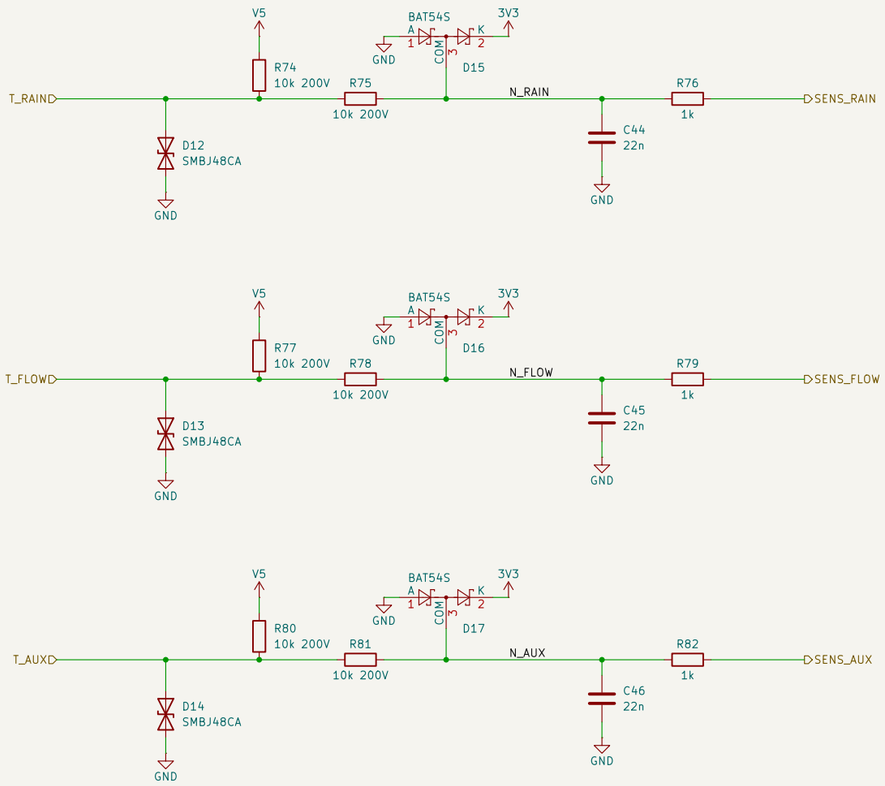
Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /USB/
File: usb.kicad_sch

Title: USB-C and power OR-ing

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/20

Three sensor inputs: RAIN, FLOW, AUX



SENSOR INPUTS (hardware design section 13).

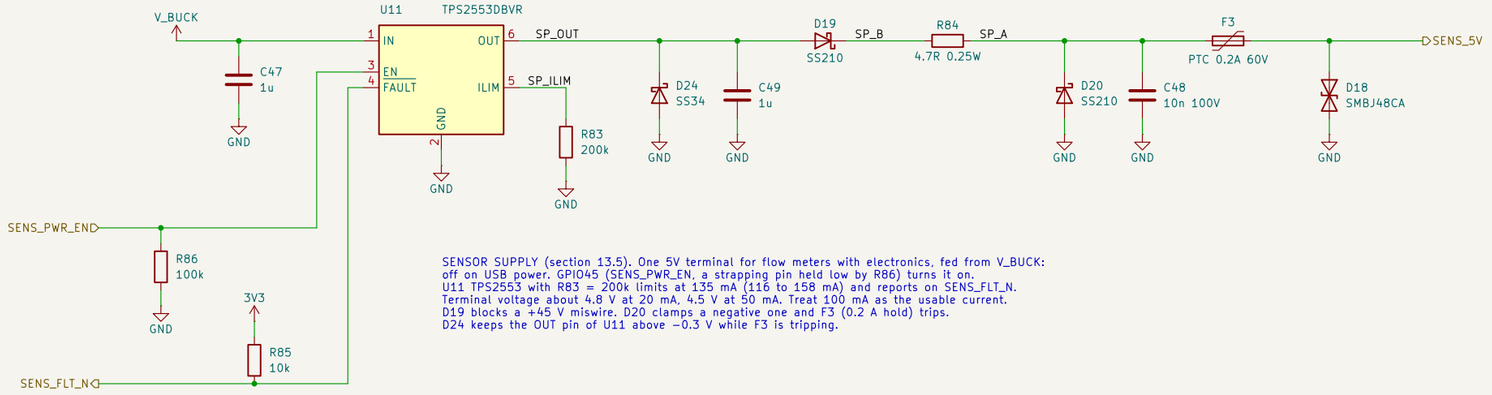
Each terminal has a 10k pull-up to V5 (0.5 mA wetting current), 10k in series, a BAT54S clamp to 3V3 and GND, 22 nF and 1k into the GPIO. A closed contact or an open-collector sensor reads low. The filter passes 200 Hz. It does not remove the 1 to 2 ms bounce of a reed-switch meter; firmware does.

32 VAC on a signal terminal for ever is survived: the TVS does not conduct below 48 V, and the pull-up and series resistors are 1206 parts rated 200 V that dissipate 0.11 W. The clamp passes 4.2 mA peak into 3V3.

D12, D13, D14, D18: SMBJ48CA at each field terminal, clamping at 77.4 V. Place each at its terminal, with a PCB spark gap where the layout allows.

The GND terminals of J4 are circuit ground, which is the transformer leg AC_N.

Switched, current-limited 5 V sensor supply



SENSOR SUPPLY (section 13.5). One 5V terminal for flow meters with electronics, fed from V_BUCK: off on USB power. GPIO45 (SENS_PWR_EN, a strapping pin held low by R86) turns it on.
U11 TPS2553 with R83 = 200k limits at 135 mA (116 to 158 mA) and reports on SENS_FLT_N.
Terminal voltage about 4.8 V at 20 mA, 4.5 V at 50 mA. Treat 100 mA as the usable current.
D19 blocks a +45 V miswire. D20 clamps a negative one and F3 (0.2 A hold) trips.
D24 keeps the OUT pin of U11 above −0.3 V while F3 is tripping.

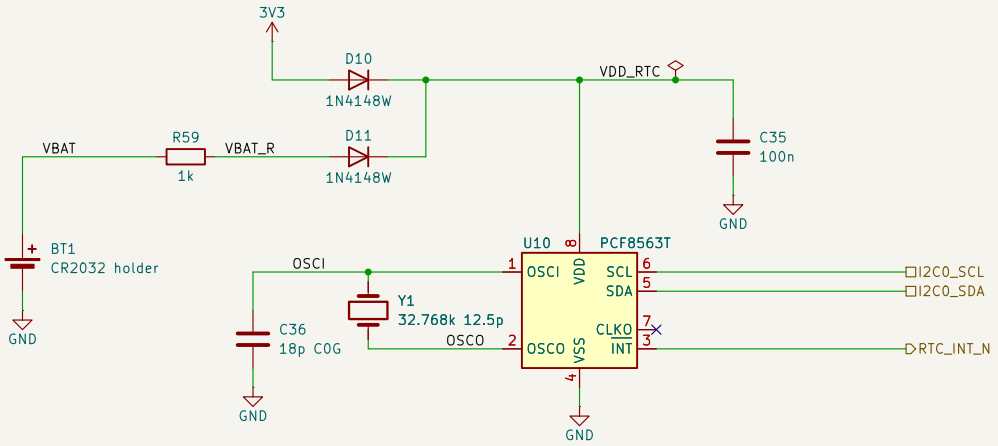
Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Sensors/
File: sensors.kicad_sch

Title: Sensor inputs and sensor supply

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 5/20

Real-time clock with coin-cell backup



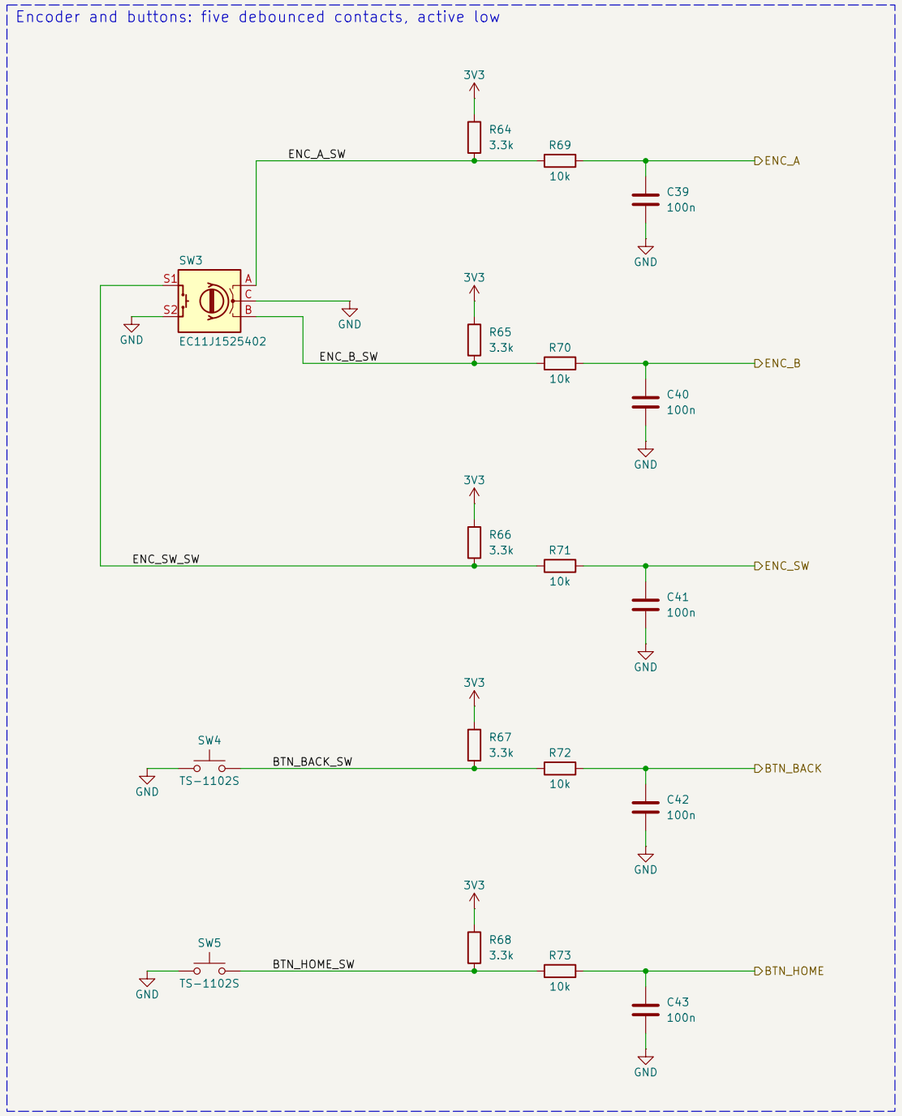
REAL-TIME CLOCK (hardware design section 14).
U10 PCF8563T at I2C address 0x51 on I2C0. It keeps time down to 1.0 V.
D10 and D11 are silicon diodes, not Schottky: a Schottky leaks as much as the RTC draws.
VDD_RTC is about 2.7 V from 3V3 and 2.6 V from the cell. R59 limits the current into the cell to under 1 mA if D11 fails short.
Y1 32.768 kHz, 12.5 pF. C36 18 pF with the 25 pF inside OSCO gives an 11.7 pF load.
CLKO is not used: firmware writes 0x00 to register 0x0D to turn the 32 kHz output off.
BT1 holds a CR2032, fitted at final assembly. Back-up current 0.25 uA typical: about ten years.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

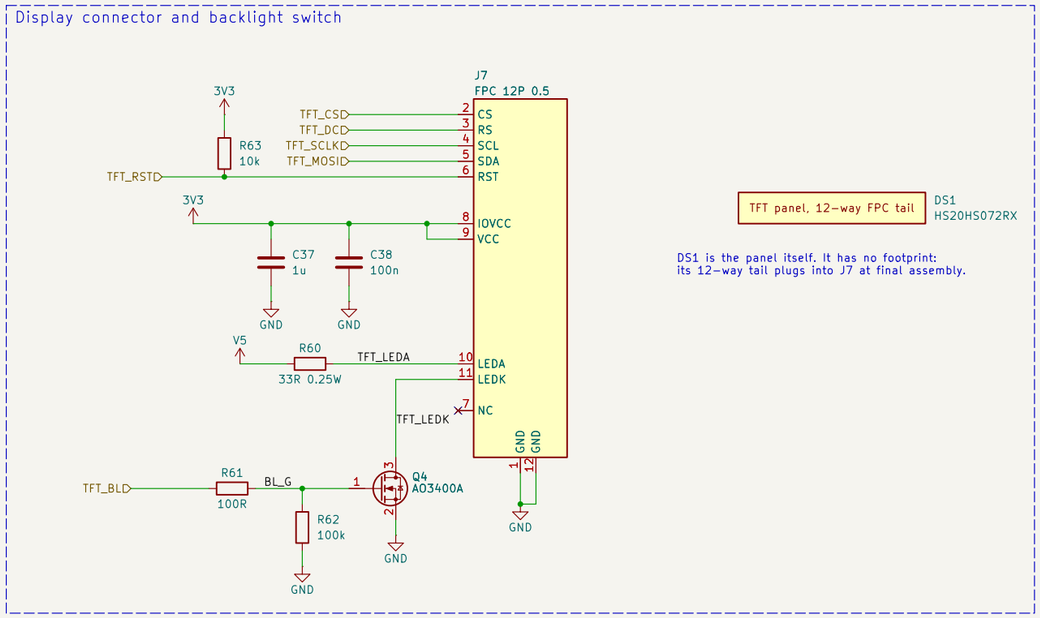
Sheet: /RTC/
File: rtc.kicad_sch

Title: Real-time clock

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 6/20



CONTROLS (section 15.4).
SW3 EC11J1525402: 30 detents, 15 pulses per turn, push switch.
SW4 Back and SW5 Manual / Home: 6 x 6 mm, 10 mm actuator.
Each contact: 3.3k pull-up (1 mA, the encoder's minimum),
then 10k and 100 nF: 1.0 ms closing, 1.3 ms opening.
All five inputs read low when the contact is closed.



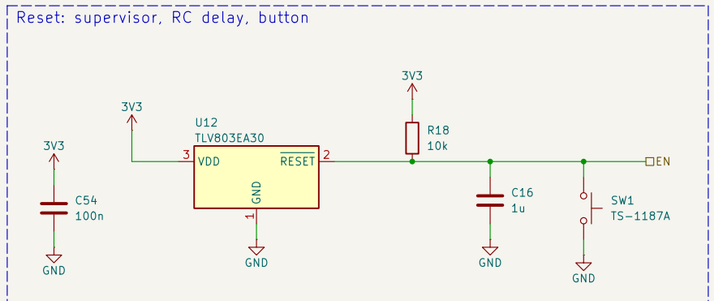
DISPLAY (hardware design section 15).
DS1 HS20HS072RX, 2.0 inch 240 x 320 IPS, ST7789, write-only 4-wire SPI on SPI2:
SCLK GPIO12, MOSI GPIO11, CS GPIO10, DC GPIO13, RST GPIO14. Start at 15 MHz.
J7 is a bottom-contact flip-lock connector. Its pad 1 is panel pin 1 (GND): the pad nearest the terminal-block edge, at y = 85.68 in the mechanical contract. The contacts of the tail are on the display face; folded under the panel they face the board. Check both on a sample panel.
Backlight: four LEDs in parallel, about 3.0 V. R60 33R from V5 gives 59 mA at 24 VAC and 27 mA at the lowest USB voltage. Q4 switches the cathode under PWM from GPIO21, 1 to 20 kHz.
R62 holds Q4 off in reset. R63 holds RST high. C37 and C38 at J7 pins 8 and 9.
LAYOUT: keep J7 within about 40 mm of module pins 18 to 22. TFT_BL keeps toggling when the firmware hangs: it must never run next to KA (section 8.5).

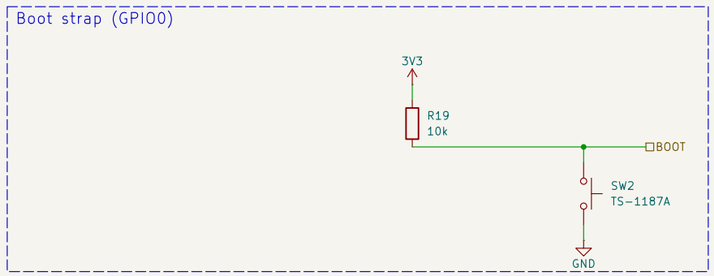
Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

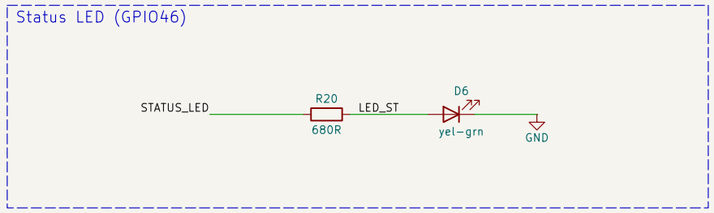
Sheet: /Display/
File: display.kicad_sch

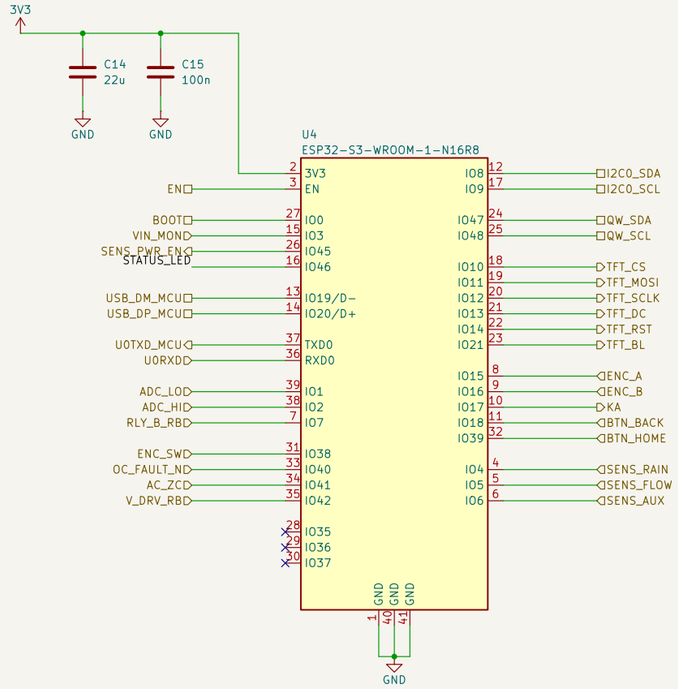
Title: Display and controls

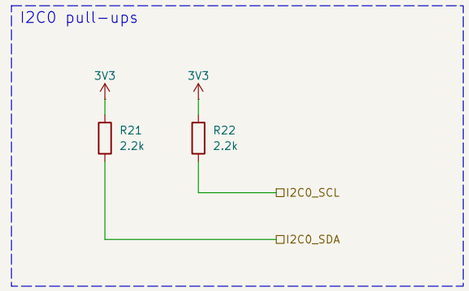
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 7/20











MCU (hardware design section 12; every pin with its reset state is in hardware/pinmap.yaml).

U4 ESP32-S3-WROOM-1-N16R8: 16 MB flash, 8 MB octal PSRAM, which takes GPIO35 to 37. Rated to 65 C, or 85 C with PSRAM ECC on: firmware must enable it. C14 and C15 at pin 2.

No valve, relay or enable function is on a GPIO. The valve outputs are on the I2C expander, which the fail-safe circuit holds in reset. Only KA (GPIO17) and I2C0 can influence a valve. KA is on module pin 10, between ENC_B and BTN_BACK, on purpose: no net beside it toggles by itself. Do not move it without redoing section 8.5 of the hardware design for its neighbours.

Strapping pins:
- GPIO0 BOOT: R19 pull-up and button SW2. Low at reset selects download mode.
- GPIO3 VIN_MON: analogue input, held low at reset by its 10k divider leg.
- GPIO45 SENS_PWR_EN: 100k to GND and one CMOS input. Never add a pull-up to this net.
- GPIO46 STATUS_LED: R20 and D6 to GND can only pull down.

ADC1 only: GPIO1 ADC_LO, GPIO2 ADC_HI, GPIO3 VIN_MON, GPIO7 RLY_B_RB. GPIO7 and GPIO42 read back the two keep-alive detectors. They stay inputs for the life of the firmware. GPIO40 reads the over-current latch through 10k. USB: GPIO19 D-, GPIO20 D+, through the series footprints on the USB sheet.

U12 TLV803EA30: open drain, pulls EN low while 3V3 is under 3.0 V (2.94 to 3.06 V) and for 130 to 270 ms after. R18 10k and C16 1 uF delay EN by about 10 ms. SW1 is reset.

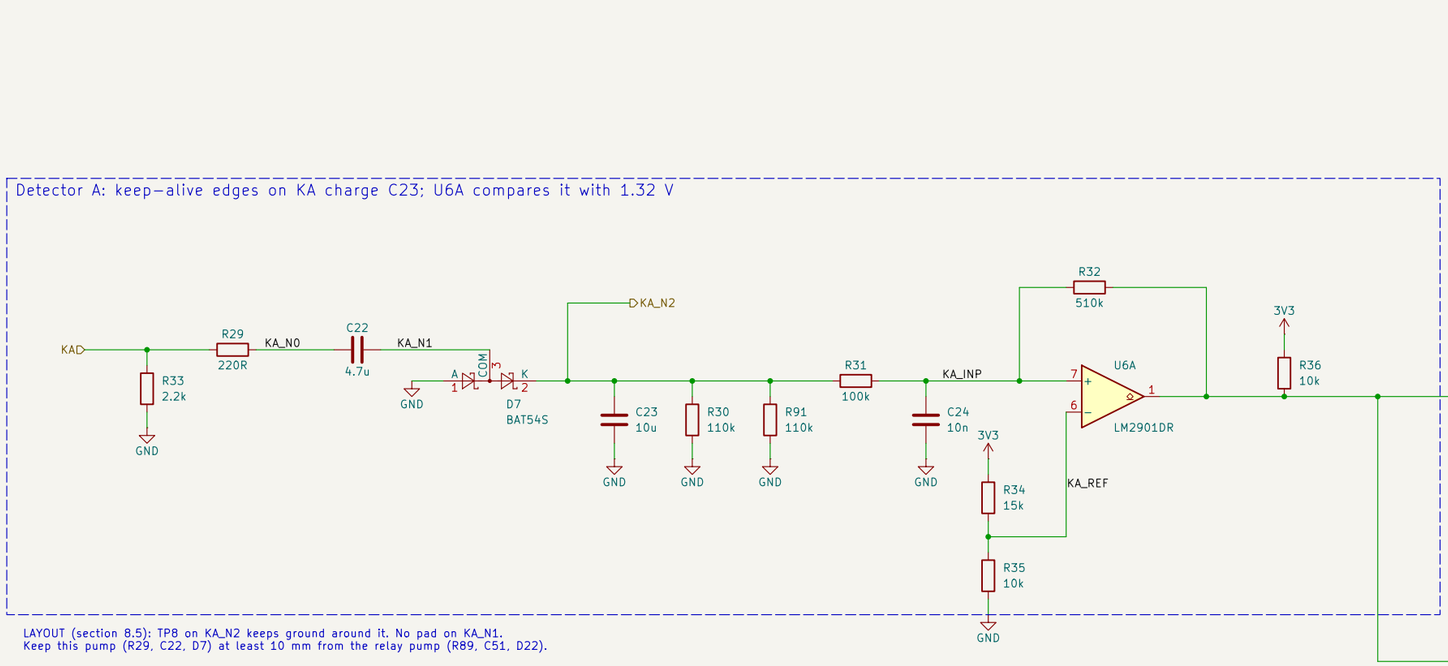
LAYOUT: the 18 x 6 mm antenna end of the module is beyond the top board edge. No copper on any layer under or beside it. Route KA from pin 10 with ground on both sides, at least 1 mm from any other signal and never next to TFT_BL, AC_ZC, U0TXD or the buck converter. Put R33 at the module pin (section 8.5).

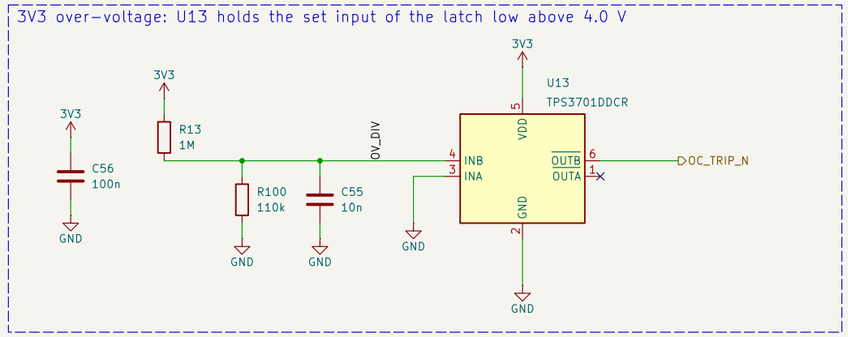
Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

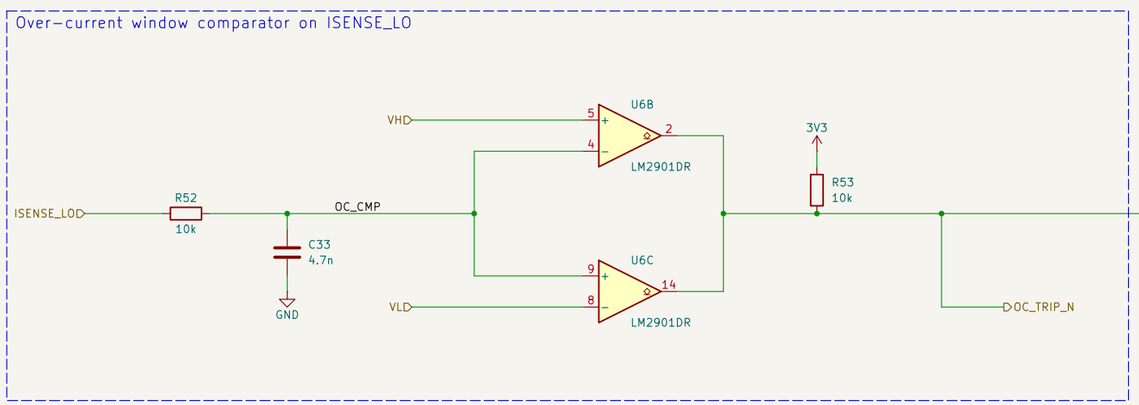
Sheet: /MCU/
File: mcu.kicad_sch

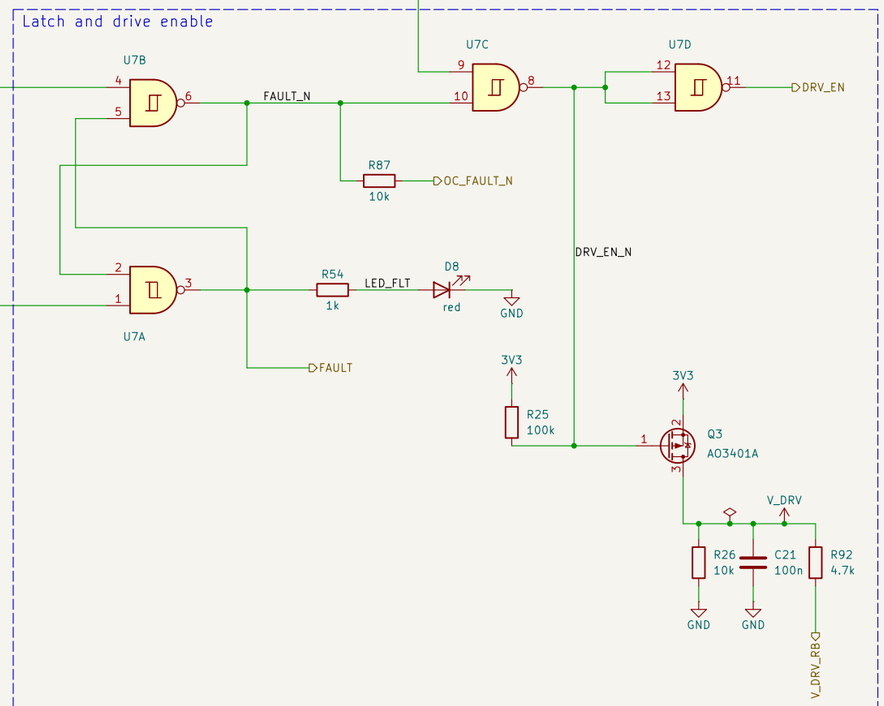
Title: MCU: ESP32-S3 module, reset, boot, status

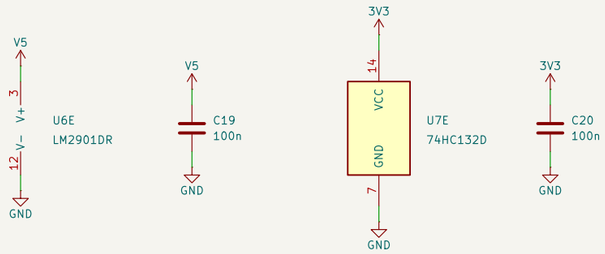
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 8/20











FAIL-SAFE GATE (hardware design section 8) and OVER-CURRENT LATCH (section 9).

Detector A. Firmware toggles KA (GPIO17) every 5 ms. It needs an edge at least every 20 ms and no phase shorter than 2.5 ms (the second is for detector B on the expander sheet). Each rising edge pushes charge through C22 and D7 into C23. R30 and R91 (two, so that one open resistor only doubles the hold-up) drain it. U6A switches on at 1.55 V on KA_N2 and off at 0.94 V (R31, R32 hysteresis; reference 1.32 V from R34, R35). With 5 ms edges V_DRV is on 13 to 22 ms after the first edge and off 0.21 to 0.83 s after the last, 1.25 s if the pin stops high (simulated corners of review A2). A steady level, high or low, never enables it. R33 holds KA low while the ESP32 is in reset. It is 2.2k so that a net behind a 10k pull-up cannot run the detectors through a solder bridge.

DRV_EN = KA_OK AND NOT FAULT. Low does three things: Q3 removes V_DRV from every opto LED, the expander is held in reset, and the expander bit that drives relay transistor Q1 is released. V_DRV_RB (through R92) is V_DRV, behind Q3, read on GPIO42: firmware times detector A with it and sees a shorted Q3 as drive that is on with no keep-alive.

Over-current. U6B pulls OC_TRIP_N low above VH (1.425 V), U6C below VL (0.188 V): 4.6 A peak in either polarity, 4.0 to 5.4 A over tolerance, about 70 us after a short begins (R52, C33: 47 us). U7A and U7B latch it. D8 lights while FAULT is set. Only KA_OK going low clears the latch. R87 stops GPIO40 from overdriving the latch node if it is ever left as an output.

U7 is a 74HC132: Schmitt inputs for the slow edges of the open-collector comparators.

3V3 over-voltage (section 9.5). If U3 shorts, 3V3 goes to V5 and the ESP32 is outside its ratings. U13 pulls OC_TRIP_N low while 3V3 is above 4.04 V (3.91 to 4.17 V): DRV_EN is then low whatever KA does, and D8 is lit. C55 with the divider is 1 ms. U13 can only remove drive.

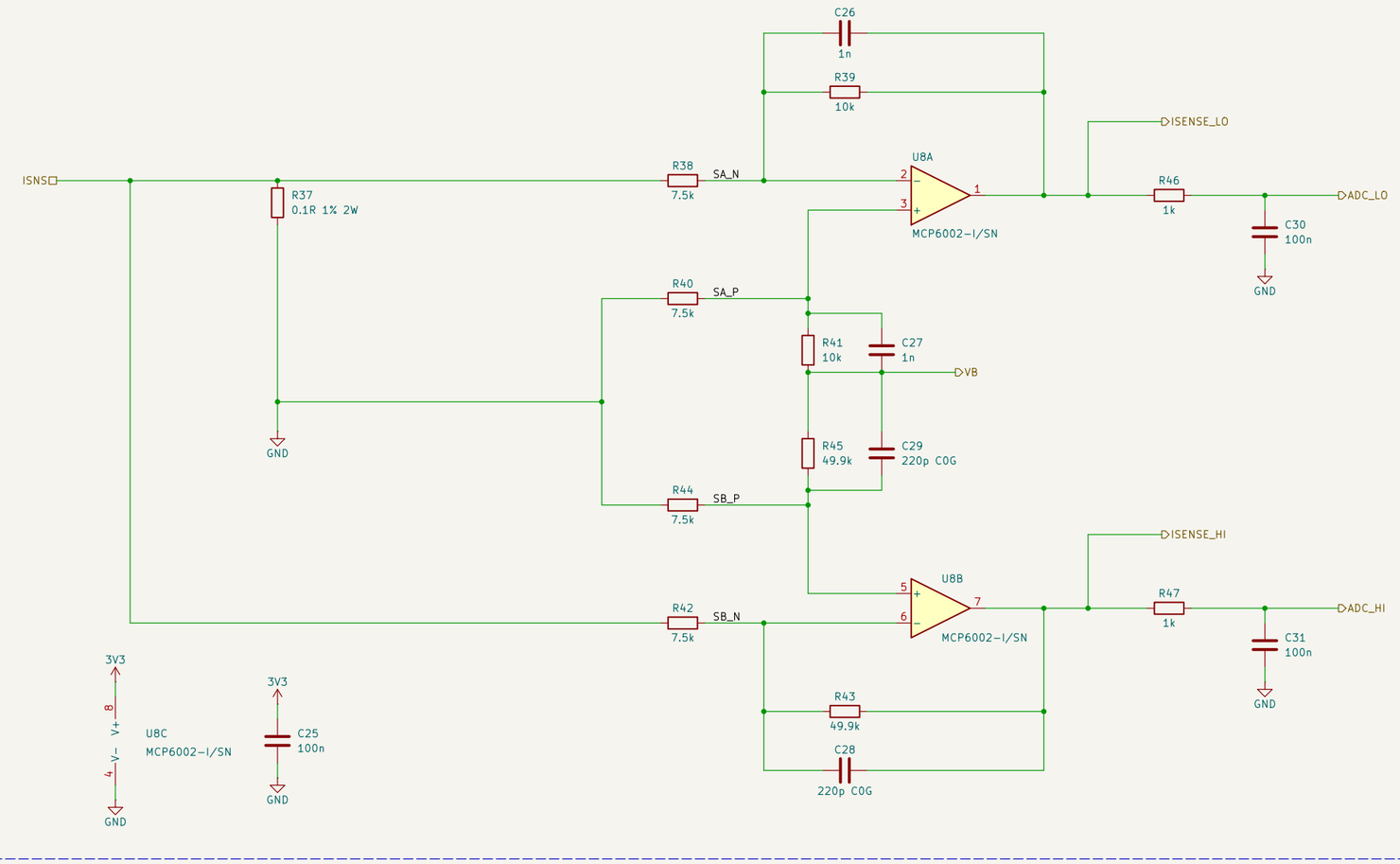
Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Failsafe/
File: failsafe.kicad_sch

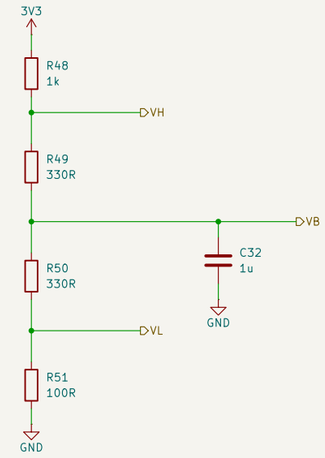
Title: Fail-safe gate and over-current latch

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 9/20

Shunt and two difference amplifiers, both biased at VB



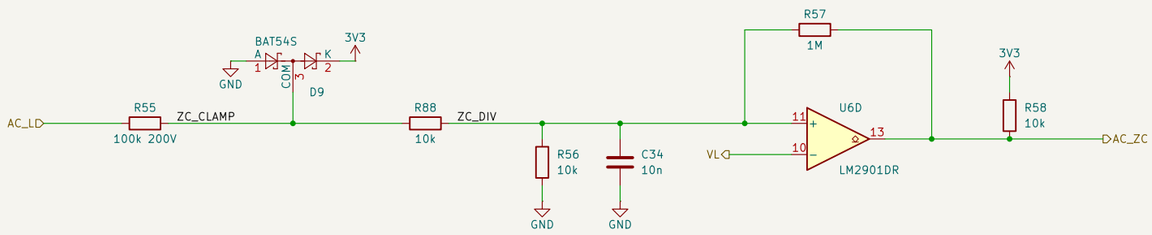
Bias and trip thresholds



CURRENT MEASUREMENT (hardware design section 10).
R37 0.1R 1 % 2 W carries the return current of every valve: ISNS is the common T1 node of all triacs.
0.23 W at 1.5 A rms. A 20 A short lifts ISNS to 2.8 V for a half-cycle; the 7.5k input resistors
keep that inside the op-amp's 2 mA input limit.
U8A: gain 10k / 7.5k = 1.333, 133.3 mV/A, linear to 5.8 A peak. U8B: 49.9k / 7.5k = 6.653,
665.3 mV/A, linear to 1.17 A peak. Both outputs are AC waveforms centred on VB, about 0.79 V.
Firmware takes the rms over 100 ms windows; there is no rectifier. C26 to C29: 15 kHz.
R46 with C30 and R47 with C31: 1.6 kHz at the ADC pins; place the capacitors at module pins 39, 38.

LAYOUT: Kelvin connections. R38 and R42 go to the ISNS pad of R37, R40 and R44 to its ground pad
with their own trace, not to the ground plane. Keep this block away from L1, the display clock
and the antenna.

Mains zero-cross and presence detector



THRESHOLDS (section 9.1). VH 1.425 V, VB 0.806 V, VL 0.188 V: trip at 0.619 V / 0.1333 V/A = 4.64 A peak.
ZERO-CROSS (section 11). R55, R88, R56 divide AC_L by 12. D9 clamps; R88 halves the clamped level so
the LM2901 input stays above −0.3 V (−0.18 V at −20 C and 32 VAC). U6D compares with VL; R57 adds
30 mV of hysteresis. AC_ZC is a square wave at mains frequency, high while AC_L is above about 2.3 V.
No edge for 25 ms means the 24 VAC is gone. R55 is a 1206 part rated 200 V.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A

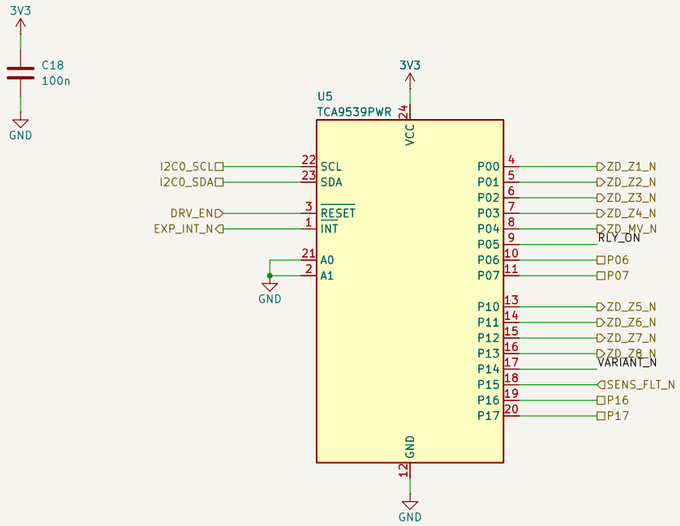
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /CurrentSense/
File: isense.kicad_sch

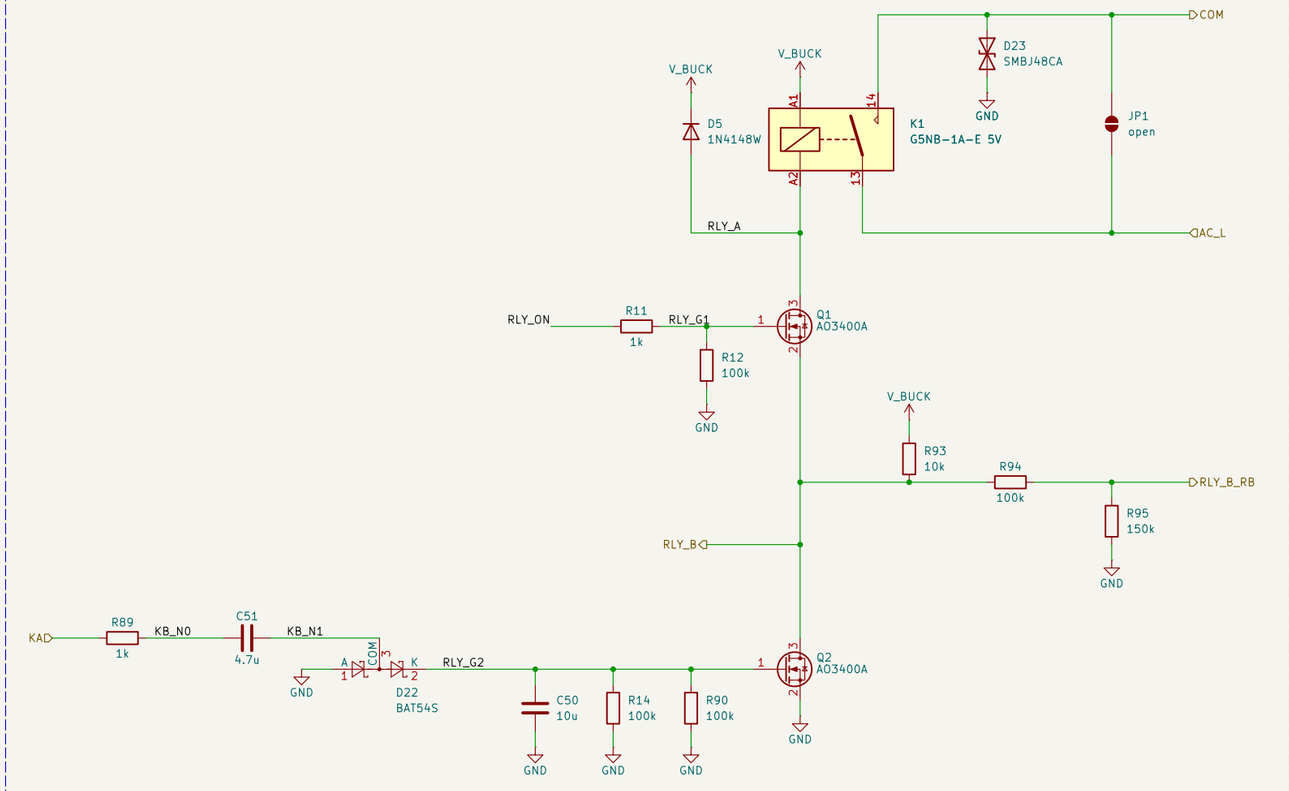
Title: Current sense and zero-cross

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 10/20

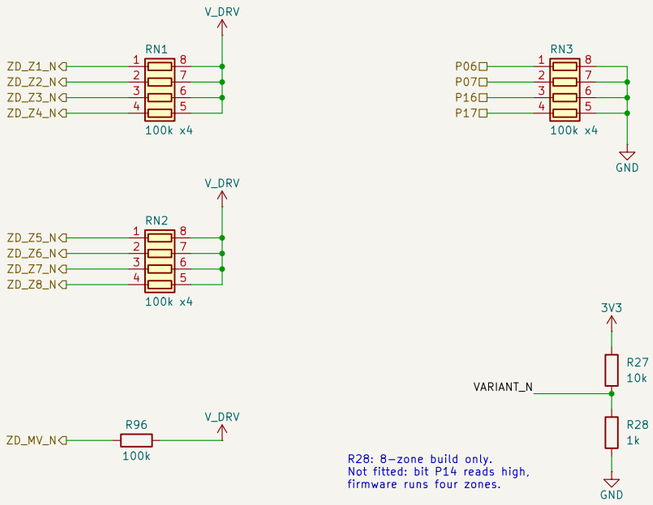
16-bit I2C expander, address 0x74, held in reset by DRV_EN



Valve-common relay: coil current passes Q1 and Q2 in series



Defined levels on every expander pin while it is in reset



VALVE-COMMON RELAY (hardware design section 7).
K1 G5NB-1A-E, 5 V 125R coil fed from V_BUCK, feeds AC_L to the COM terminals. With K1 open no valve has power, whatever a shorted triac, TVS or opto-triac does. It closes only when Q1 AND Q2 conduct.
Q1: on while expander bit RLY_ON is high. DRV_EN low resets the expander and R12 turns Q1 off in 0.1 ms.
Q2: on while the relay detector sees edges on KA. R89, C51, D22 pump the reservoir C50; R14 and R90 (two, so that one open resistor only doubles the time) drain it. Gate about 2.7 V while toggling. Q2 lets go 0.10 to 1.45 s after the last edge, within 2.19 s if the pin stops high (simulated corners of review A2; the long end is a low-threshold Q2 when hot).
R93, R94, R95: read-back of the node between Q1 and Q2 on GPIO7. 3.05 V = Q2 off, 0 V = Q2 on.
D23 SMBJ48CA clamps COM while K1 is open and takes the inductive kick when it opens under load.
JP1: open. Bridge only if the relay is not fitted (then the fault analysis of section 8.5 is void).
LAYOUT: no test pad on RLY_G2, KB_N1 or KB_N0. Surround the gate node of Q2 with ground: a leakage path from any rail to that node is the only way to hold Q2 on. Keep this pump (R89, C51, D22) at least 10 mm from the gate-drive pump (R29, C22, D7), and turn C51 at right angles to C22.
K1 is through-hole, inserted from the bottom side.

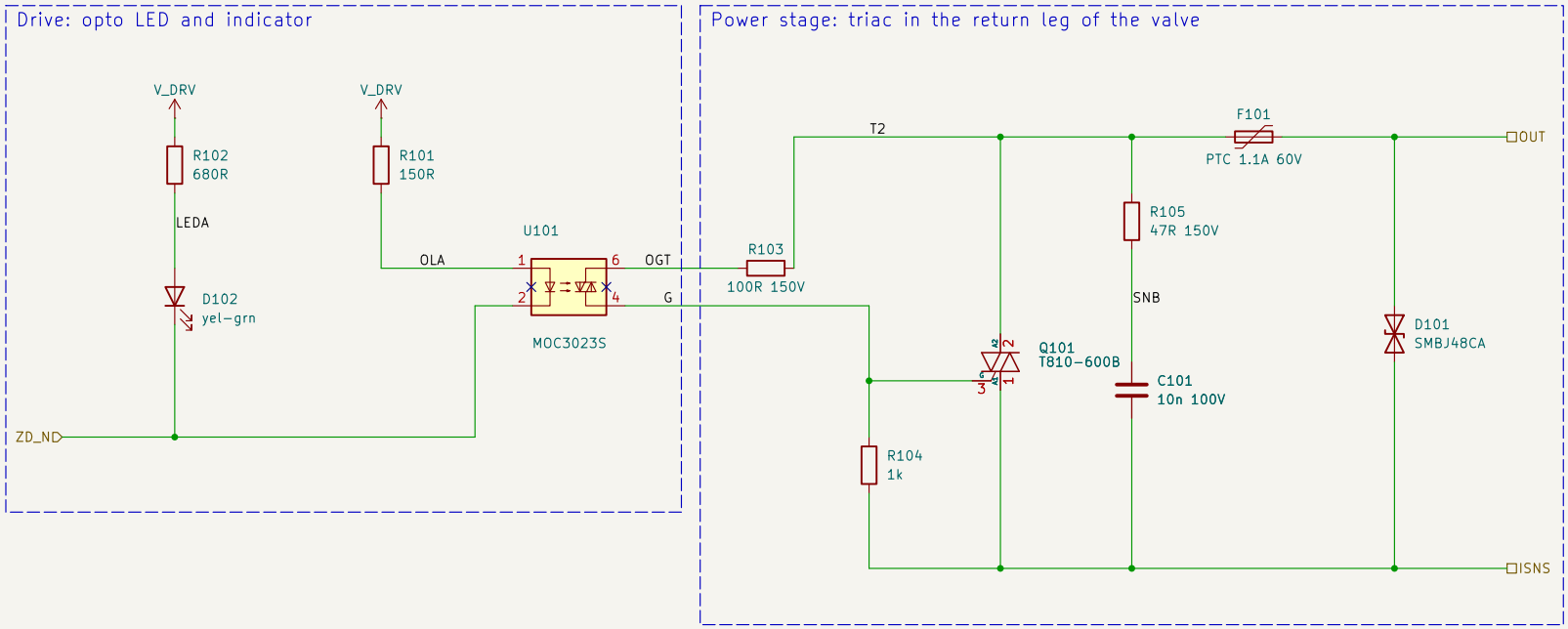
EXPANDER (section 8.2). U5 TCA9539: every pin is an input while RESET is low, which is whenever no valve runs. A valve is on when its bit is an output and low: P00 to P03 Z1 to Z4, P04 MV, P10 to P13 Z5 to Z8. P05 RLY_ON is on when high. P14 reads the build variant, P15 the sensor supply fault.
RN1, RN2, R96 hold the valve pins at V_DRV, not 3V3, so an off opto LED never sees reverse voltage.
RN3 holds the four spare pins low. RN2 is fitted on both builds.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Expander/
File: expander.kicad_sch

Title: I/O expander and valve-common relay

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 11/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
The valve sits between a COM terminal and OUT. Current returns through PTC F1, the triac
and the shared shunt (ISNS) to GND. The gate is fired from T2 through opto-triac U1 and R3,
so gate current has the polarity of T2: quadrants I and III, which every triac supports.

Q1 T810-600B-TR, 8 A, three-quadrant, DPAK. Pin 1 A1 (T1) to ISNS, pin 2 and tab A2 (T2), pin 3 gate.
    The tab is net T2: give it its own copper island of 2 cm2 or more (section 6.7).
U1 MOC3023, random phase. LED current 7.1 mA worst case, 12 mA typical; it needs 5.6 mA at -20 C.
R1 150R sets the opto LED current. R2 680R gives the indicator LED 1.6 mA.
R3 100R limits the opto-triac to 0.45 A at the 45 V peak. R4 1k holds the gate at T1.
R5 + C1 47R and 10 nF snubber. F1 PTC 1.1 A hold, 40 A maximum. D1 SMBJ48CA clamps at 77 V.
D2 lights when the expander pin is low AND V_DRV is on: the real drive state.

ZD_N is low to turn the output on. V_DRV is removed by the fail-safe circuit.

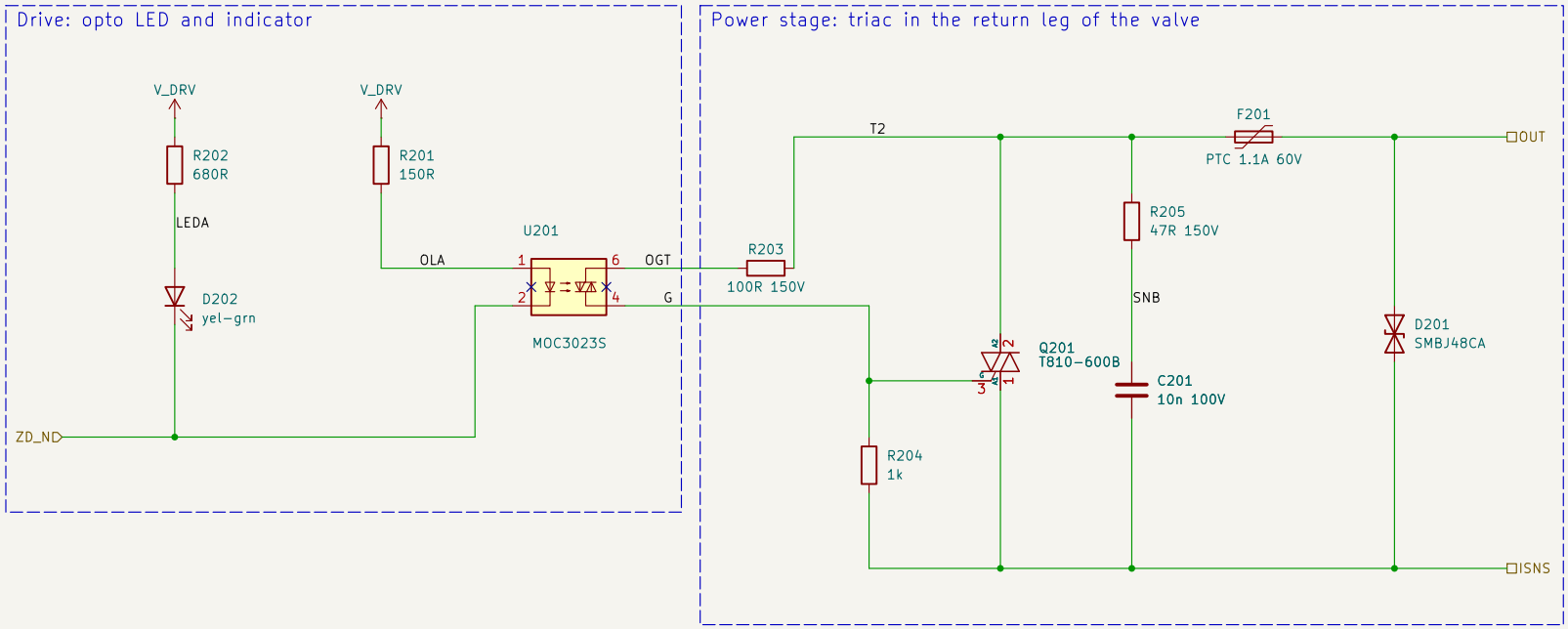
4-ZONE BUILD: instances Z5 to Z8 are not fitted
(KiCad variant "4-zone"). Z1 to Z4 and MV always are.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z1/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 12/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
The valve sits between a COM terminal and OUT. Current returns through PTC F1, the triac
and the shared shunt (ISNS) to GND. The gate is fired from T2 through opto-triac U1 and R3,
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R3 100R limits the opto-triac to 0.45 A at the 45 V peak. R4 1k holds the gate at T1.
R5 + C1 47R and 10 nF snubber. F1 PTC 1.1 A hold, 40 A maximum. D1 SMBJ48CA clamps at 77 V.
D2 lights when the expander pin is low AND V_DRV is on: the real drive state.

ZD_N is low to turn the output on. V_DRV is removed by the fail-safe circuit.

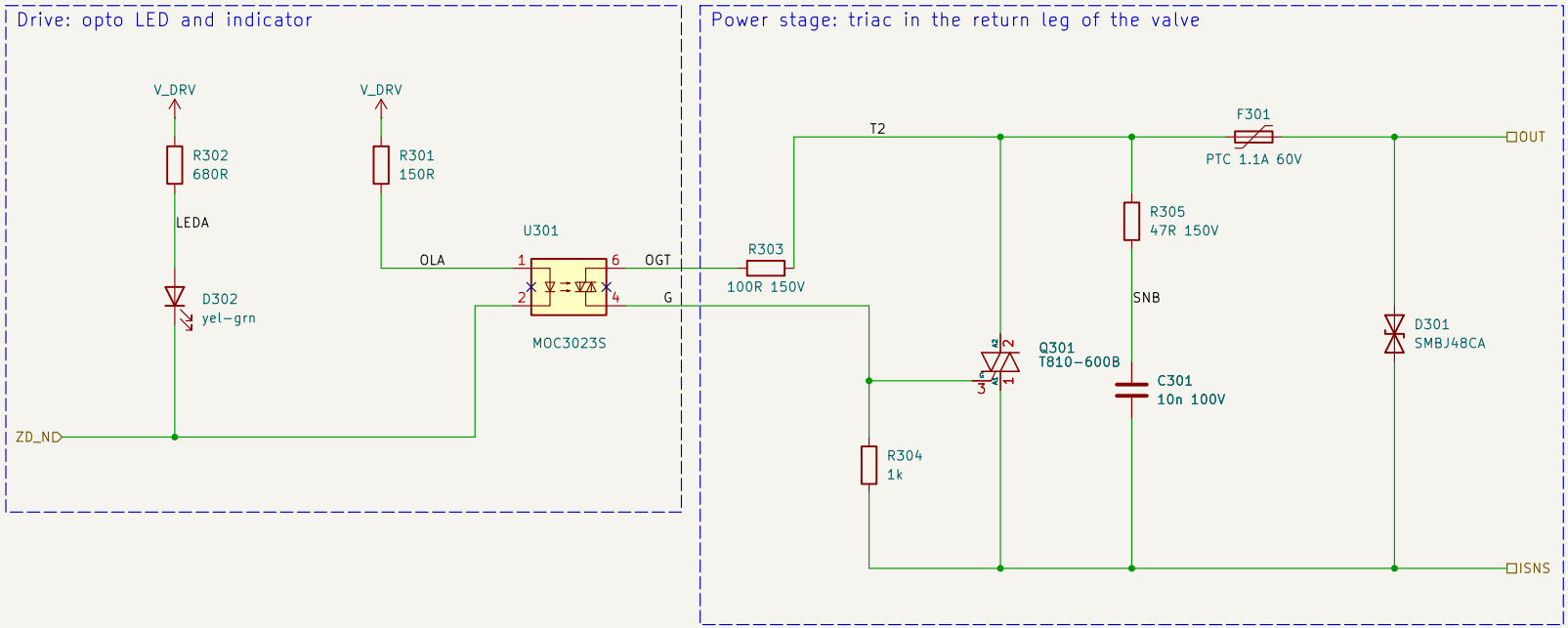
4-ZONE BUILD: instances Z5 to Z8 are not fitted
(KiCad variant "4-zone"). Z1 to Z4 and MV always are.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z2/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 13/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
The valve sits between a COM terminal and OUT. Current returns through PTC F1, the triac
and the shared shunt (ISNS) to GND. The gate is fired from T2 through opto-triac U1 and R3,
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R5 + C1 47R and 10 nF snubber. F1 PTC 1.1 A hold, 40 A maximum. D1 SMBJ48CA clamps at 77 V.
D2 lights when the expander pin is low AND V_DRV is on: the real drive state.

ZD_N is low to turn the output on. V_DRV is removed by the fail-safe circuit.

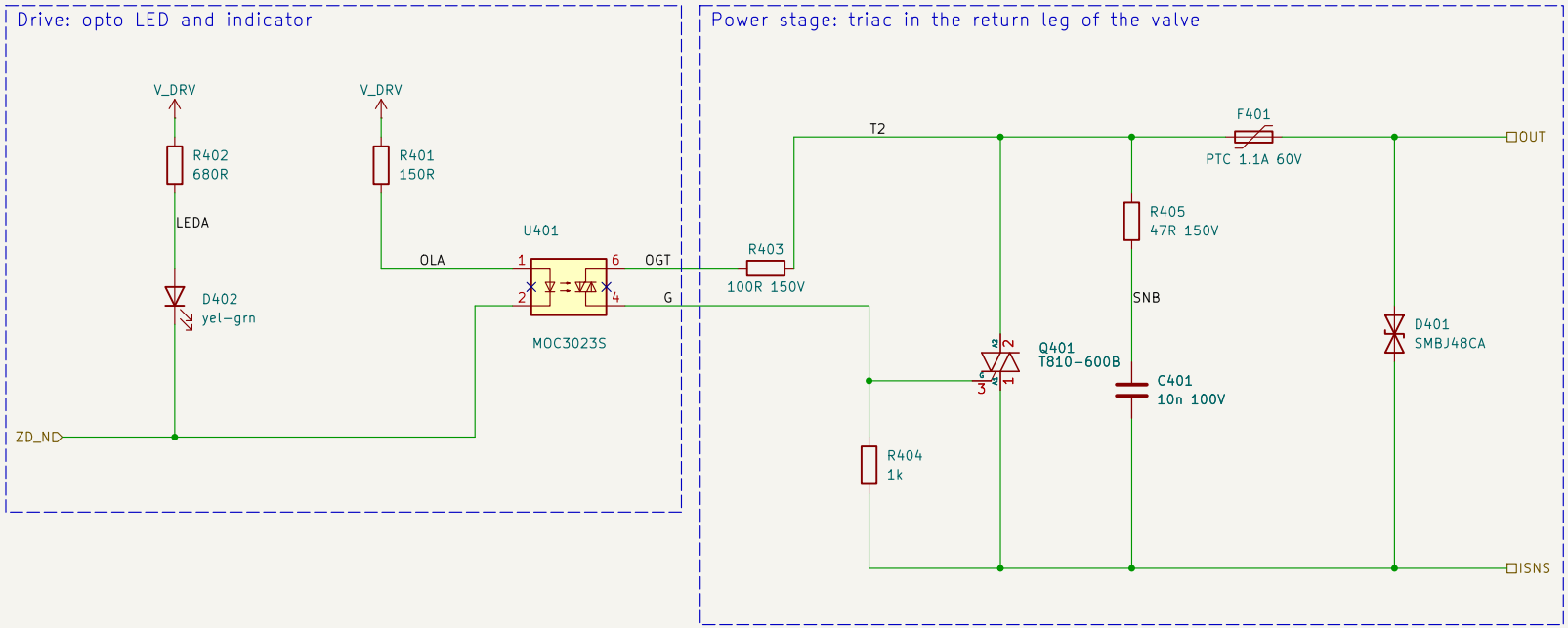
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Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z3/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 14/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
The valve sits between a COM terminal and OUT. Current returns through PTC F1, the triac
and the shared shunt (ISNS) to GND. The gate is fired from T2 through opto-triac U1 and R3,
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R5 + C1 47R and 10 nF snubber. F1 PTC 1.1 A hold, 40 A maximum. D1 SMBJ48CA clamps at 77 V.
D2 lights when the expander pin is low AND V_DRV is on: the real drive state.

ZD_N is low to turn the output on. V_DRV is removed by the fail-safe circuit.

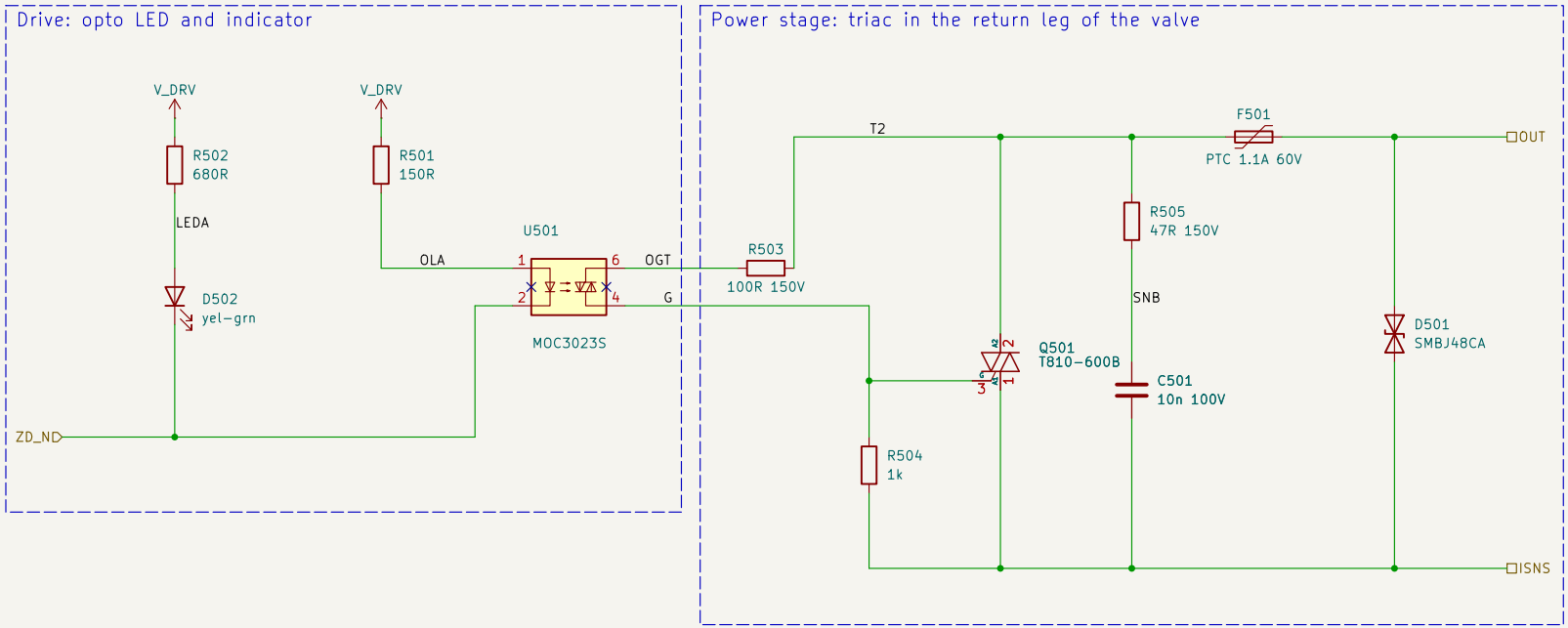
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Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z4/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 15/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
The valve sits between a COM terminal and OUT. Current returns through PTC F1, the triac
and the shared shunt (ISNS) to GND. The gate is fired from T2 through opto-triac U1 and R3,
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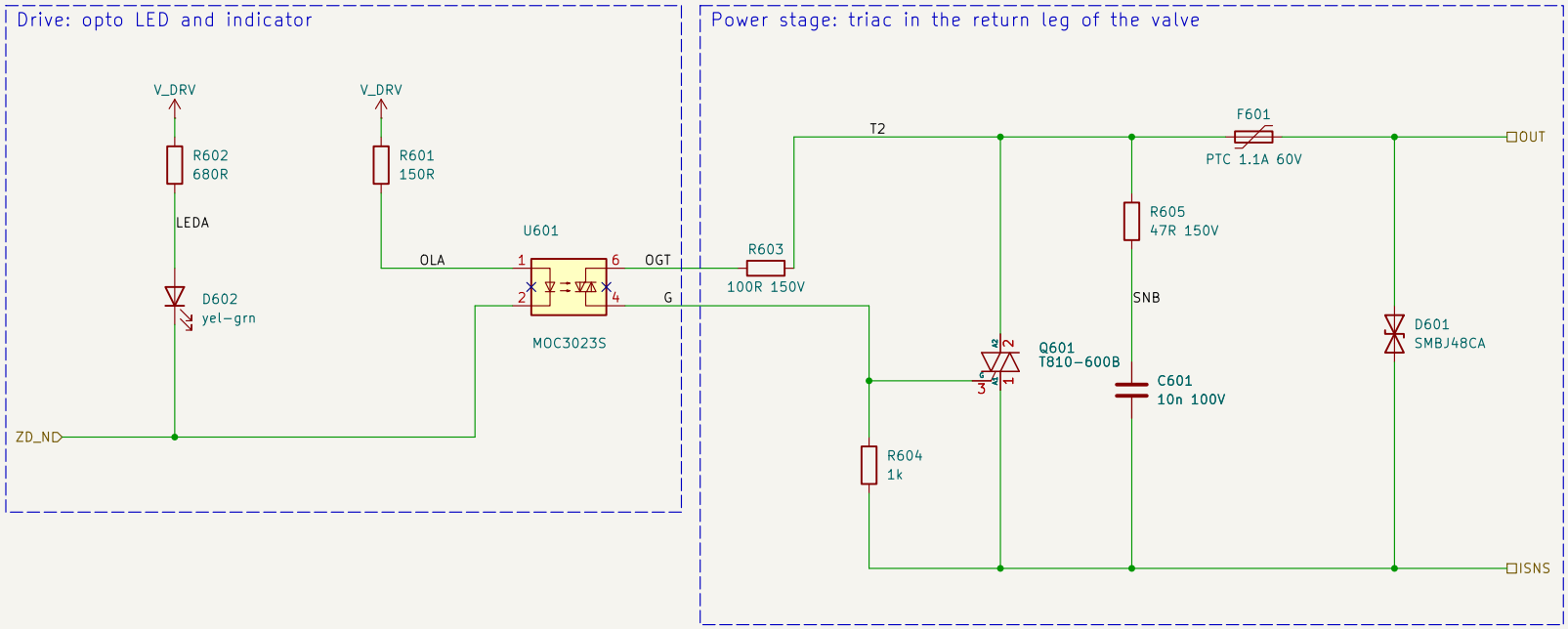
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Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z5/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 16/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
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D2 lights when the expander pin is low AND V_DRV is on: the real drive state.

ZD_N is low to turn the output on. V_DRV is removed by the fail-safe circuit.

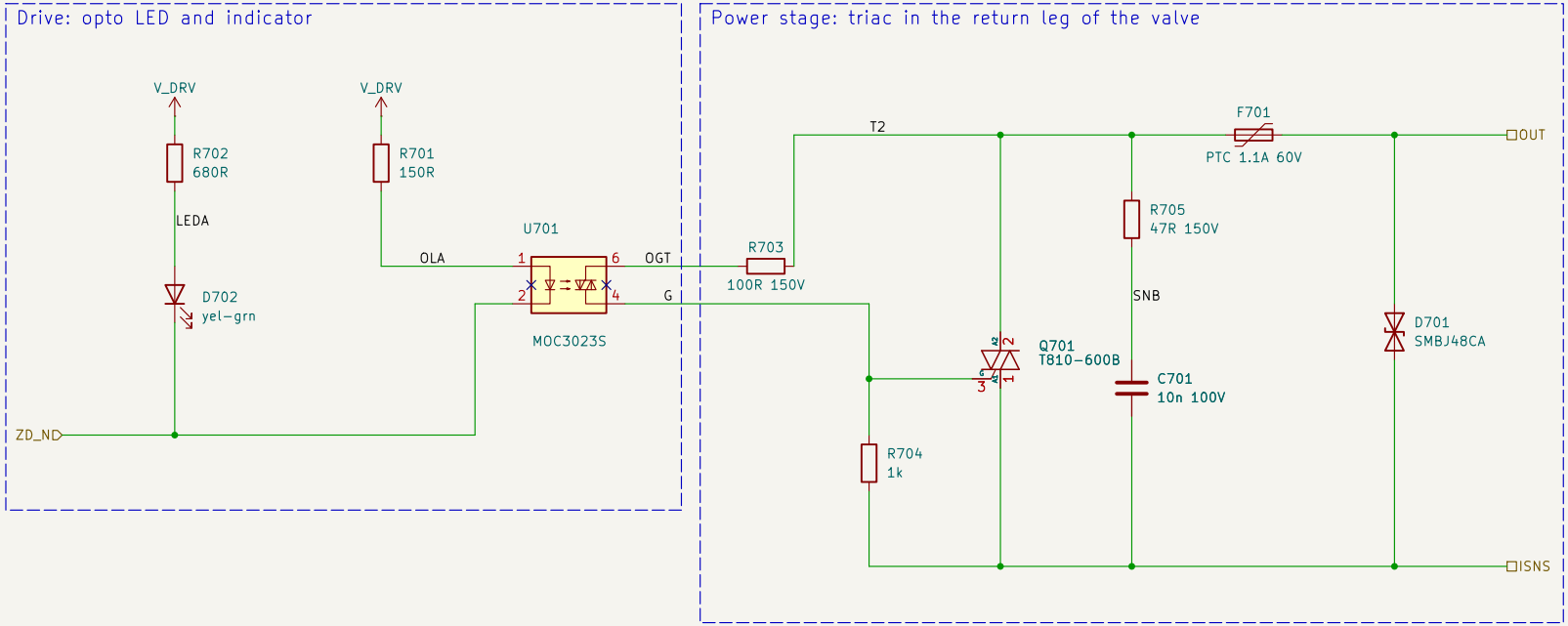
4-ZONE BUILD: instances Z5 to Z8 are not fitted
(KiCad variant "4-zone"). Z1 to Z4 and MV always are.

Generated from design/circuit.py and hardware/netlist.yaml. Do not edit by hand.
Paper design. Never built, nothing measured.
ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z6/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 17/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
The valve sits between a COM terminal and OUT. Current returns through PTC F1, the triac
and the shared shunt (ISNS) to GND. The gate is fired from T2 through opto-triac U1 and R3,
so gate current has the polarity of T2: quadrants I and III, which every triac supports.

Q1 T810-600B-TR, 8 A, three-quadrant, DPAK. Pin 1 A1 (T1) to ISNS, pin 2 and tab A2 (T2), pin 3 gate.
    The tab is net T2: give it its own copper island of 2 cm2 or more (section 6.7).
U1 MOC3023, random phase. LED current 7.1 mA worst case, 12 mA typical; it needs 5.6 mA at -20 C.
R1 150R sets the opto LED current. R2 680R gives the indicator LED 1.6 mA.
R3 100R limits the opto-triac to 0.45 A at the 45 V peak. R4 1k holds the gate at T1.
R5 + C1 47R and 10 nF snubber. F1 PTC 1.1 A hold, 40 A maximum. D1 SMBJ48CA clamps at 77 V.
D2 lights when the expander pin is low AND V_DRV is on: the real drive state.

ZD_N is low to turn the output on. V_DRV is removed by the fail-safe circuit.

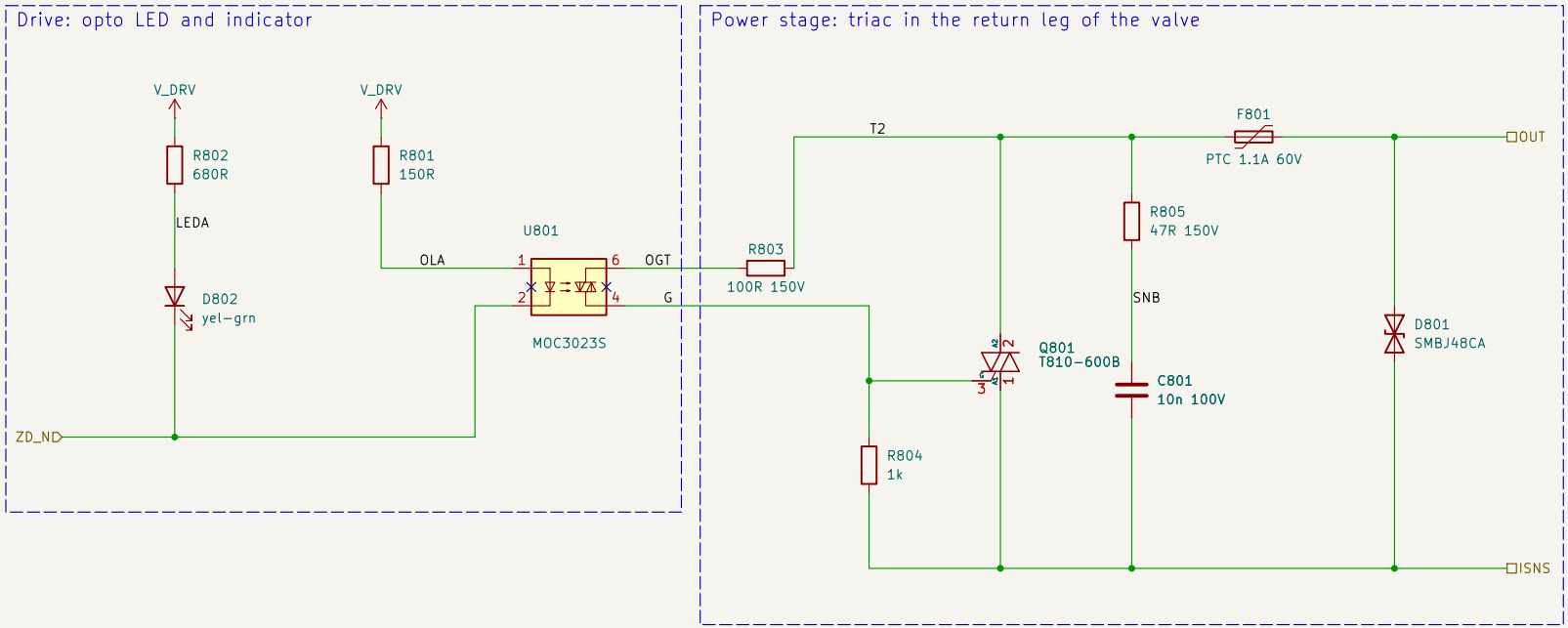
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ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z7/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 18/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
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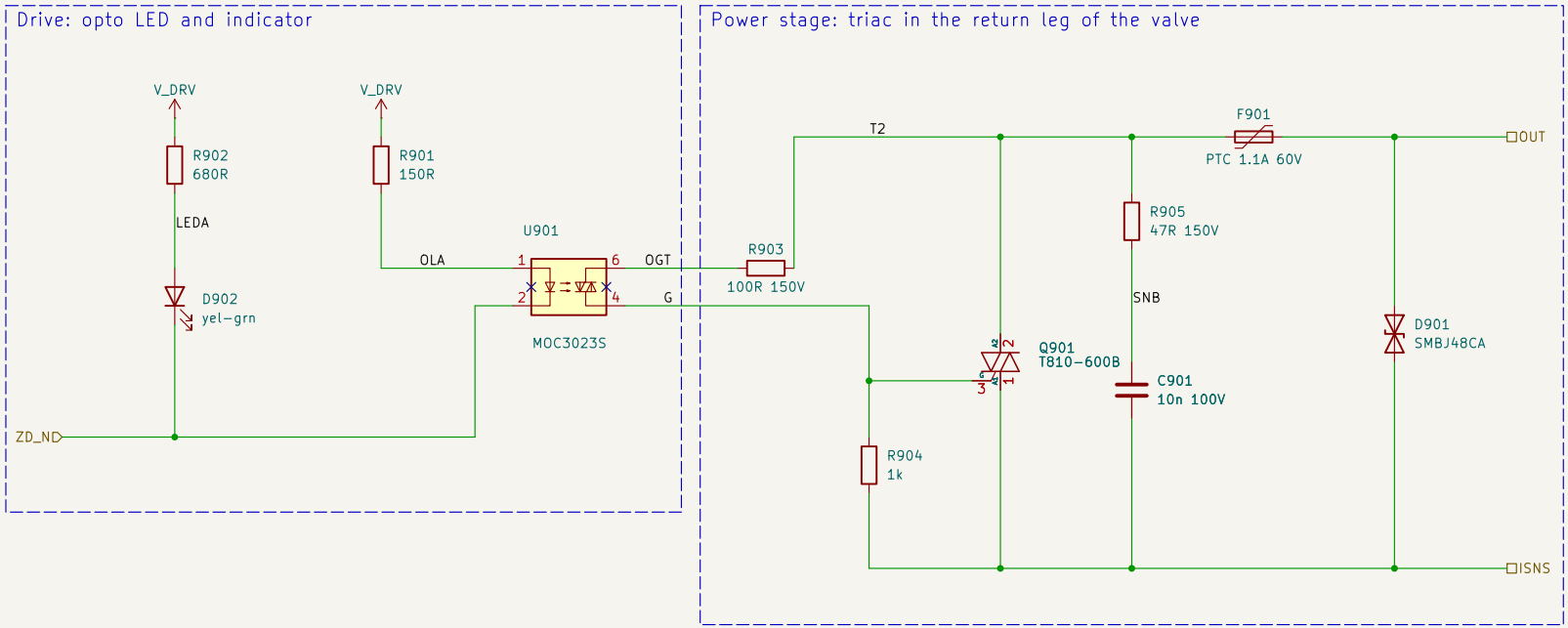
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ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /Z8/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 19/20



OUTPUT CHANNEL (hardware design section 6). Nine instances: Z1 to Z8 and MV.
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ZoneFlow irrigation controller, revision A
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /MV/
File: channel.kicad_sch

Title: 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 20/20